Motor Imagery EEG classification using spectral feature and Siamese neural network

Github URL https://github.com/scholar-lab/C_Neuro_project
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Introduction

The project comprises of the following steps.

1. Power Spectral Density Feature (PSD) Extraction.
2. Training Siamese Neural Network to classify a pair of input as similar or different.
3. Evaluation of the model performance using the trained Model.

Power Spectral Density Feature Extraction

BCI 2008 2a dataset is used for the project. The four class motor imagery (MI) EEG is used to generate PSD samples. The time series plot of a subject is as shown in Figure 1. The plot of the mean of the PSD samples obtained is as shown in Figure 2.

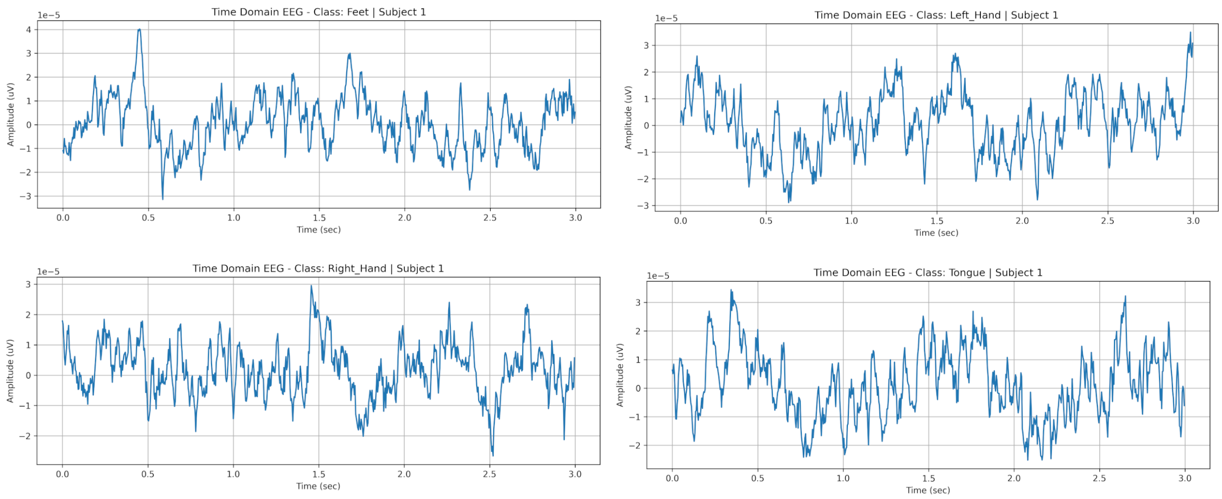

Figure 1: Time Domain plot of a subject EEG.

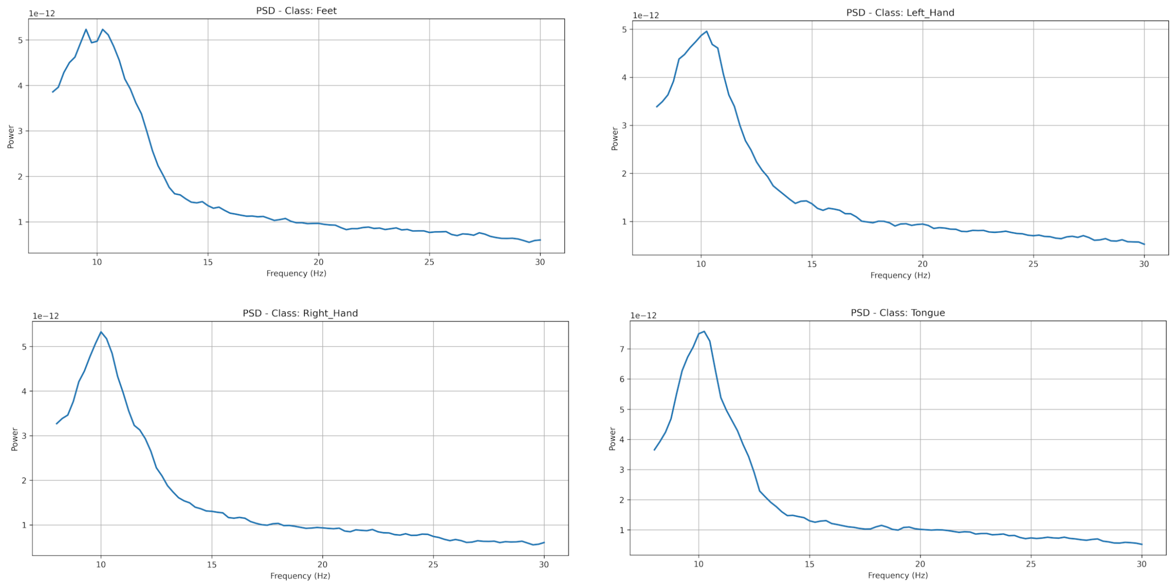

Figure 2: Mean PSD of each class EEG.

Training the using Siamese Neural Network

The PSD feature is used to train the the Siamese Network using contrastive loss. The training and testing process is as shown in Figure 3.

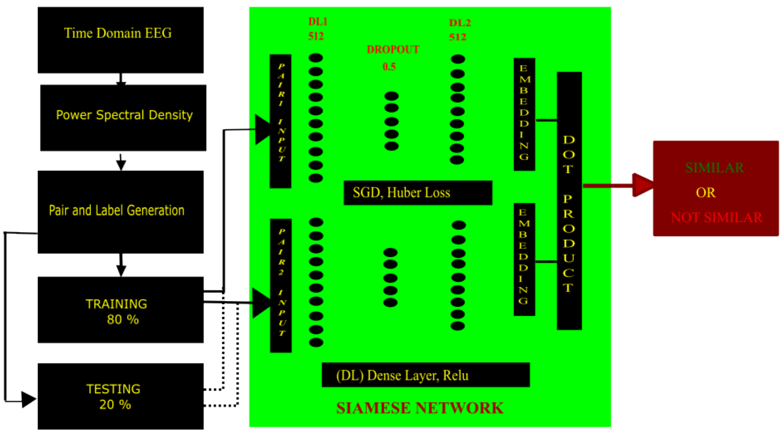

Figure 3: Siamese Neural Network Architecture

Evaluation

The test accuracy is 92.96%. The confusion matrix for the test data is as shown in Figure 4. The training accuracy and loss curve is as shown in Figure 5 and Figure 6 respectively. The datapoint of multiple epoch in a loop are averaged and the resulting data points are binned in a group of 200 to make the plot smoother.

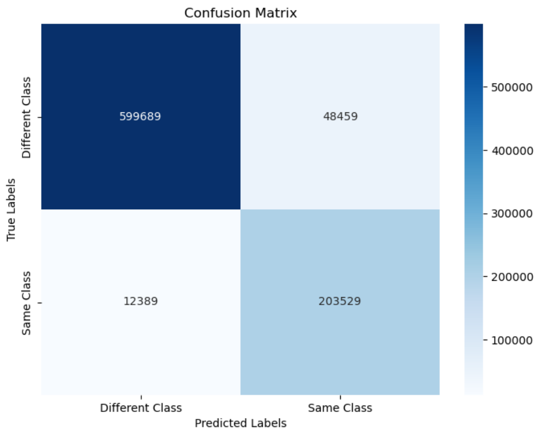

Figure 4: Confusion Matrix

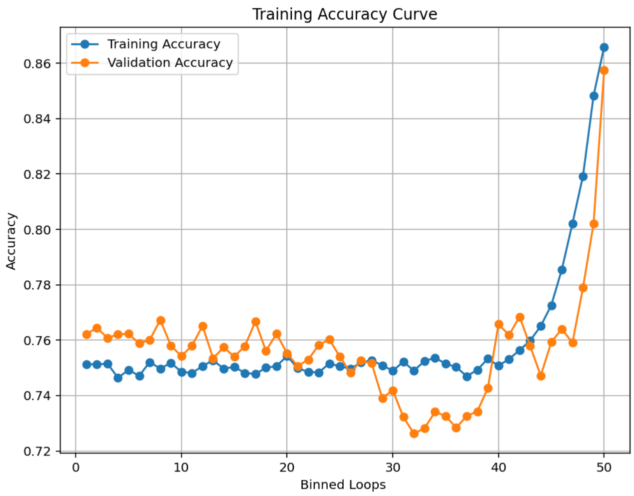

Figure 5: Training Accuracy plot

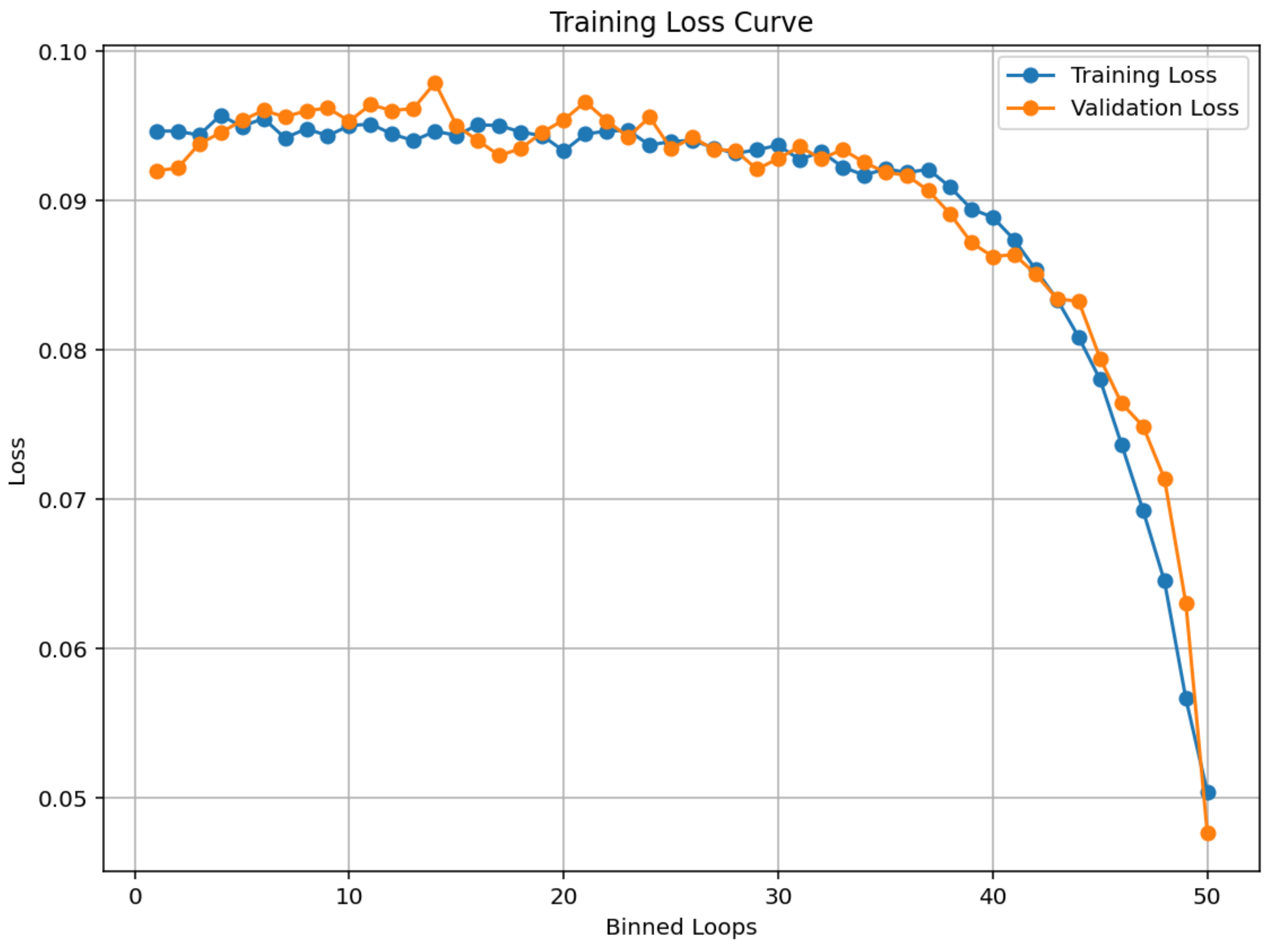

Figure 6: Training Loss plot